



Software Engineering - SET 1

Section 1: Basics of Software Engineering

1. Which of the following is not a software engineering activity?

- a) Design
- b) Testing
- c) Documentation
- d) Compiling

Answer: d) Compiling

Explanation: Compiling is a part of the programming process, not a software engineering activity.

2. The primary aim of software engineering is to:

- a) Reduce hardware cost
- b) Increase programming skill
- c) Produce reliable software
- d) Make documentation easy

Answer: c) Produce reliable software

Explanation: Software engineering focuses on producing reliable and maintainable software systems.

3. Which one of these is not a software process model?

- a) Waterfall
- b) Spiral
- c) Prototyping
- d) Compilation

Answer: d) Compilation

Explanation: Compilation is not a software process model; the others are standard SDLC models.

4. Which model is also known as the linear sequential model?

- a) Waterfall



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- b) Spiral
- c) Agile
- d) RAD

Answer: a) Waterfall

Explanation: The Waterfall model is a linear sequential design approach.

5. Which of the following is not a phase of SDLC?

- a) Requirement gathering
- b) Testing
- c) Coding
- d) Deployment

Answer: c) Coding

Explanation: 'Implementation' is the proper SDLC term, though coding happens during it.

Section 2: Software Development Models

6. In the Spiral model, what does each loop represent?

- a) A phase
- b) A prototype
- c) A risk
- d) A bug

Answer: a) A phase

Explanation: Each loop in the spiral model represents a phase in the development process.

7. Which model is best suited for small projects with clear requirements?

- a) Spiral
- b) Agile
- c) Waterfall
- d) V-Model



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Answer: c) Waterfall

Explanation: The Waterfall model is ideal when requirements are clearly defined upfront.

8. **V-Model is an extension of which SDLC model?**

- a) Spiral
- b) Waterfall
- c) Agile
- d) RAD

Answer: b) Waterfall

Explanation: V-Model is a refinement of the waterfall model with testing mapped to each phase.

9. **Which model is best suited when requirements change frequently?**

- a) Agile
- b) Waterfall
- c) V-Model
- d) Spiral

Answer: a) Agile

Explanation: Agile is iterative and handles frequent requirement changes efficiently.

10. **Which model emphasizes risk analysis?**

- a) Waterfall
- b) Spiral
- c) Agile
- d) Prototyping

Answer: b) Spiral

Explanation: Risk analysis is a key component of the spiral model.

Section 3: Software Requirements & Design

11. **Which document outlines what the system should do?**

- a) SDD
- b) SRS



- c) TDD
- d) DFD

Answer: b) SRS

Explanation: Software Requirement Specification (SRS) defines system functionalities.

12. Functional requirements define:

- a) Performance and usability
- b) External behavior of the system
- c) Constraints of design
- d) Hardware specifications

Answer: b) External behavior of the system

Explanation: Functional requirements describe what the system should do.

13. Which of these is a tool used in structured design?

- a) Flowchart
- b) UML
- c) DFD
- d) Sequence Diagram

Answer: c) DFD

Explanation: DFD (Data Flow Diagram) shows the flow of data in a structured design.

14. What does UML stand for?

- a) Unified Module Language
- b) Universal Modeling Language
- c) Unified Modeling Language
- d) Useful Modeling Language

Answer: c) Unified Modeling Language

Explanation: UML is a standardized modeling language used in software design.

15. Which UML diagram represents object interaction over time?

- a) Use case diagram
- b) Class diagram
- c) Sequence diagram
- d) State diagram



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Answer: c) Sequence diagram

Explanation: Sequence diagrams show object interaction in a time sequence.

Section 4: Testing and Quality Assurance

16. Which testing verifies that the system works as expected?

- a) System testing
- b) Alpha testing
- c) Unit testing
- d) Functional testing

Answer: d) Functional testing

Explanation: Functional testing checks whether the system performs its intended functions.

17. Which testing is done by end users at the developer's site?

- a) Unit testing
- b) Beta testing
- c) Alpha testing
- d) Integration testing

Answer: c) Alpha testing

Explanation: Alpha testing is done by users at the development site before product release.

18. Which testing focuses on individual units of code?

- a) Integration testing
- b) Unit testing
- c) Regression testing
- d) System testing

Answer: b) Unit testing

Explanation: Unit testing targets the smallest testable parts of an application.

19. Which of the following is not a functional testing type?

- a) Smoke testing



- b) Load testing
- c) Regression testing
- d) Sanity testing

Answer: b) Load testing

Explanation: Load testing is a type of non-functional performance testing.

20. Which testing technique is based on code coverage?

- a) Black box
- b) Regression
- c) White box
- d) Integration

Answer: c) White box

Explanation: White box testing requires internal knowledge of code and logic.

Section 5: Software Maintenance & Metrics

21. What type of maintenance is done to adapt the software to a new environment?

- a) Corrective
- b) Adaptive
- c) Preventive
- d) Perfective

Answer: b) Adaptive

Explanation: Adaptive maintenance modifies software to adapt to new environments.

22. Which software metric measures lines of code?

- a) Function Point
- b) LOC
- c) Cyclomatic complexity
- d) Test coverage

Answer: b) LOC

Explanation: LOC (Lines of Code) is a basic metric for code size.



23. Which of the following is not a software metric?

- a) Cyclomatic complexity
- b) Coupling
- c) Abstraction
- d) Cohesion

Answer: c) Abstraction

Explanation: Abstraction is a design principle, not a measurable software metric.

24. Which is not a valid type of maintenance?

- a) Adaptive
- b) Corrective
- c) Predictive
- d) Preventive

Answer: c) Predictive

Explanation: Predictive is not a recognized software maintenance type.

25. Cyclomatic complexity is used to measure:

- a) Design errors
- b) Number of modules
- c) Logical complexity
- d) Test case count

Answer: c) Logical complexity

Explanation: Cyclomatic complexity measures the decision structure complexity.

Section 6: Software Project Management

26. What is the purpose of a Gantt chart in project management?

- a) Cost analysis
- b) Time scheduling
- c) Testing results
- d) Code quality



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Answer: b) Time scheduling

Explanation: Gantt charts visually represent the project schedule.

27. Which of these is not a software project management activity?

- a) Project planning
- b) Risk management
- c) Software testing
- d) Scheduling

Answer: c) Software testing

Explanation: Testing is a development activity, not project management itself.

28. Which of the following is a risk management activity?

- a) Estimation
- b) Mitigation
- c) Design
- d) Scheduling

Answer: b) Mitigation

Explanation: Risk mitigation is a strategy to reduce the impact of risks.

29. PERT is used to:

- a) Estimate project cost
- b) Measure productivity
- c) Evaluate time and activity flow
- d) Check software reliability

Answer: c) Evaluate time and activity flow

Explanation: PERT (Program Evaluation and Review Technique) is for scheduling and planning.

30. The term 'baseline' in project management refers to:

- a) A project deadline
- b) Original scope or plan
- c) Coding standard
- d) Project report

Answer: b) Original scope or plan

Explanation: A baseline is the reference point against which progress is measured.



Section 7: Miscellaneous & Conceptual Questions

31. **Software is a product and a:**

- a) Service
- b) Theory
- c) Hardware
- d) Memory

Answer: a) Service

Explanation: Software can be sold as a service (SaaS).

32. **Software engineering principles include all except:**

- a) Maintainability
- b) Reliability
- c) Manufacturability
- d) Usability

Answer: c) Manufacturability

Explanation: Manufacturability applies to physical products, not software.

33. **Which of the following is a CASE tool?**

- a) MS Word
- b) MS Excel
- c) Rational Rose
- d) Windows OS

Answer: c) Rational Rose

Explanation: Rational Rose is a CASE (Computer-Aided Software Engineering) tool.

34. **Which activity ensures the software meets quality standards?**

- a) Verification
- b) Validation
- c) Quality assurance
- d) Maintenance

Answer: c) Quality assurance

Explanation: QA is the process that ensures quality standards are met.



35. Which one is not a layer of software architecture?

- a) Presentation layer
- b) Application layer
- c) Database layer
- d) Hardware layer

Answer: d) Hardware layer

Explanation: Hardware is not part of software architecture.

36. A feasibility study is done to:

- a) Design software
- b) Develop SRS
- c) Assess project viability
- d) Start implementation

Answer: c) Assess project viability

Explanation: Feasibility studies determine whether a project is worth pursuing.

37. What is the full form of RAD?

- a) Rapid Application Development
- b) Reliable App Design
- c) Robust Application Design
- d) Random Access Development

Answer: a) Rapid Application Development

Explanation: RAD is a fast-paced software development methodology.

38. Which is not a black box testing type?

- a) Boundary value analysis
- b) Equivalence partitioning
- c) Condition testing
- d) Decision table testing

Answer: c) Condition testing

Explanation: Condition testing is a white-box testing technique.

39. Which diagram is used to represent workflows in software design?

- a) Class diagram
- b) Activity diagram



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- c) Use case diagram
- d) ER diagram

Answer: b) Activity diagram

Explanation: Activity diagrams represent workflows and control flows.

40. What is the purpose of regression testing?

- a) Test new features
- b) Check UI layout
- c) Ensure old features still work after changes
- d) Measure performance

Answer: c) Ensure old features still work after changes

Explanation: Regression testing verifies that recent changes haven't broken existing functionality.